```
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error
np.random.seed(42)
```

```
n = 150
locations = ['Urban', 'Suburban', 'Rural']
df = pd.DataFrame({
 'area_sqft': np.random.randint(250, 5000, n),
 'num_rooms': np.random.randint(2, 5, n),
 'location': np.random.choice(locations, n),
 'age_years': np.random.randint(2, 30, n),
})
```

```
loc_map = {'Urban': 1.5, 'Suburban': 1.0, 'Rural': 0.7}#expensiv medium cheap
noise = np.random.normal(0, 30000, n)
df['price'] = (
 df['area_sqft'] * 160 +
 df['num_rooms'] * 45000 -
 df['age_years'] * 1000 +
 df['location'].map(loc_map) * 50000 +
 noise
).astype(int)
```

```
print(df.head(8).to_string(index=False))
print(f"\nDataset shape : {df.shape}")
print(df.describe()[['area_sqft','num_rooms','age_years','price']].round(0))
```

```
 area_sqft  num_rooms location  age_years  price
      3455          3    Rural          4 704396
      1064          3 Suburban         19 340074
       595          4    Rural         19 302213
      2599          3    Urban         18 592065
       612          4    Rural         25 283988
      2693          2 Suburban          4 505822
      2515          3    Urban          4 630363
      2438          4 Suburban          7 645482

Dataset shape : (150, 5)
       area_sqft  num_rooms  age_years     price
count      150.0      150.0      150.0     150.0
mean      2742.0        3.0       15.0  610204.0
std       1318.0        1.0        8.0  209204.0
min        304.0        2.0        2.0  199519.0
25%       1566.0        2.0        8.0  425298.0
50%       2974.0        3.0       14.0  646408.0
75%       3740.0        4.0       23.0  778445.0
max       4938.0        4.0       29.0  961788.0
```

```
le = LabelEncoder()
```

```
df['location_enc'] = le.fit_transform(df['location'])
```

```
features = ['area_sqft', 'num_rooms', 'age_years', 'location_enc']
X = df[features]
y = df['price']
```

```
print(df[['location', 'location_enc']].drop_duplicates().sort_values('location_enc'))
print(f"\nFeature matrix shape : {X.shape}")
print(f"Target vector shape : {y.shape}")
```

```
   location  location_enc
0     Rural             0
6  Suburban             1
1     Urban             2

Feature matrix shape : (200, 4)
Target vector shape : (200,)
```

```
X_train, X_test, y_train, y_test = train_test_split(
 X, y, test_size=0.2, random_state=42
)
```

```
print(f"Training samples : {len(X_train)} ({len(X_train)/len(X)*100:.0f}%)")
print(f"Testing samples : {len(X_test)} ({len(X_test)/len(X)*100:.0f}%)")
print(f"Feature names : {features}")
```

```
Training samples : 160 (80%)
Testing samples : 40 (20%)
Feature names : ['area_sqft', 'num_rooms', 'age_years', 'location_enc']
```

```
model = LinearRegression()
model.fit(X_train, y_train)
print("Learned Coefficients:")
print("-" * 45)
for feat, coef in zip(features, model.coef_):
 direction = "↑" if coef > 0 else "↓"
 print(f" {feat:15s}: Rs {coef:>10,.0f} {direction}")
print(f" {'intercept':15s}: Rs {model.intercept_:>10,.0f}")
print("-" * 45)
```

```
Learned Coefficients:
---------------------------------------------
 area_sqft      : Rs        151 ↑
 num_rooms      : Rs     23,169 ↑
 age_years      : Rs     -2,262 ↓
 location_enc   : Rs     18,114 ↑
 intercept      : Rs     46,789
---------------------------------------------
```

```
y_pred = model.predict(X_test)
r2 = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
```

```
print("=" * 40)
print(" MODEL EVALUATION REPORT")
print("=" * 40)
print(f"R² Score : {r2:.4f} {'✓ Excellent' if r2>0.9 else '✓ Good' if r2>0.8 else '■ Needs work'}")
print(f"MAE : Rs {mae:>10,.0f}")
print(f"RMSE : Rs {rmse:>10,.0f}")
print("=" * 40)
```

```
========================================
 MODEL EVALUATION REPORT
========================================
R² Score : 0.9872 ✓ Excellent
MAE : Rs     13,439
RMSE : Rs     16,959
========================================
```

```
print("\nSample Predictions:")
print(f"{'Actual':>10} | {'Predicted':>10} | {'Error':>10}")
print("-" * 36)
for a, p in zip(y_test[:8], y_pred[:8]):
 print(f"{a:>10,} | {p:>10,.0f} | {p-a:>+10,.0f}")
```

```

Sample Predictions:
    Actual |  Predicted |      Error
------------------------------------
   489,899 |    478,320 |    -11,579
   601,329 |    572,877 |    -28,452
   623,788 |    650,489 |    +26,701
   291,458 |    271,086 |    -20,372
   390,040 |    344,431 |    -45,609
   123,884 |    134,545 |    +10,661
   484,442 |    443,295 |    -41,147
   562,823 |    556,350 |     -6,473
```

```
new_houses = pd.DataFrame({
 'area_sqft': [1300, 1800, 1450, 4000],
 'num_rooms': [4, 4, 2, 7 ],
 'age_years': [6, 11, 45, 5 ],
 'location_enc':[3, 2, 1, 5],
 'location_lbl':['Urban','Suburban','Rural','Urban'],
})
```

```
predictions = model.predict(new_houses[features])
print("\n HOUSE PRICE PREDICTIONS")
print("=" * 50)
```

```

 HOUSE PRICE PREDICTIONS
```

```
==================================================
```

```
for i, (_, row) in enumerate(new_houses.iterrows()):
 print(f" House {i+1}: {row.area_sqft} sqft | "
 f"{row.num_rooms} rooms | "
 f"{row.age_years}yr old | "
 f"{row.location_lbl}")
 print(f" Estimated Price → Rs {predictions[i]:>10,.0f}")
 print()
```

```
 House 1: 1300 sqft | 4 rooms | 6yr old | Urban
 Estimated Price → Rs    375,993

 House 2: 1800 sqft | 4 rooms | 11yr old | Suburban
 Estimated Price → Rs    421,861

 House 3: 1450 sqft | 2 rooms | 45yr old | Rural
 Estimated Price → Rs    227,792

 House 4: 4000 sqft | 7 rooms | 5yr old | Urban
 Estimated Price → Rs    890,569
```